

Spectral Silence and Geometric Chaos in an Exactly Degenerate Topological Manifold

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Exact degeneracy removes the internal energy spacings on which conventional random-matrix diagnostics of quantum chaos rely. We show that the projector geometry of the same manifold can remain statistically nontrivial. For a $D = 50$ bosonic fractional-quantum-Hall zero-mode manifold, the raw energy spectral form factor is the featureless constant $K_{E,\text{raw}} = D$, and its connected part vanishes identically. By contrast, the spectrum of the quantum-metric-normalized non-Abelian Berry curvature has a connected correlation ramp quantitatively described by the finite- D complex Jacobi ensemble. This agreement is not a one-point-density artifact: a momentum-resolved structured tangent family has the same active rank but strongly violates local repulsion, the ramp, and long-range number variance. Two interventions then separate the diagnoses causally. A Poisson-to-GUE interpolation inside a fixed projector changes the energy gap ratio from 0.385645 to 0.598677 without changing the curvature spectrum, whereas scrambling the projector-to-complement tangent channels produces Jacobi local statistics at $g = 0.20$ and a registered ramp window only at $g = 0.40$. We derive the unfolded finite-rank Jacobi form factor from its determinantal kernel and prove that boundary atoms at rank $D > M$ suppress the connected plateau exactly to k/D , with $k = 2M - D$. The physical ramp is universal over the tested Fourier window, while number variance becomes nonuniversal beyond $L = 1.0$. Thus Berry curvature does not merely replace one spectrum by another: it reveals a statistically independent hierarchy of quantum-geometric correlations inside an energy-silent topological manifold.

I. INTRODUCTION

Quantum chaos is conventionally diagnosed in the spectrum of a Hamiltonian. After resolving symmetries and unfolding the density of states, chaotic levels repel and their long-range correlations approach a Wigner-Dyson random-matrix ensemble [1, 2]. The connected spectral form factor packages the same two-level correlations into a ramp and plateau. The eigenstate thermalization hypothesis (ETH) complements this spectral picture by describing matrix elements of observables in the energy basis [3–5]. All three constructions use nonzero energy differences.

An exactly degenerate manifold is therefore a sharp failure mode, not merely an inconvenient finite-size limit. If D states share the energy E_0 , then

$$\begin{aligned} Z_E(t) &= \sum_{a=1}^D e^{-itE_a} = D e^{-itE_0}, \\ K_{E,\text{raw}}(t) &= \frac{|Z_E(t)|^2}{D} = D, \end{aligned} \quad (1)$$

and the ensemble-connected form factor is identically zero. There are no spacings to unfold and no ramp to recover. Adding an arbitrary splitting may manufacture an energy statistic, but it changes the problem and can obscure the structure responsible for the degeneracy.

The alternative studied here is geometric. An isolated degenerate eigenspace defines a vector bundle over Hamiltonian parameter space. Its adiabatic transport is governed by the Wilczek–Zee connection and the matrix-valued Berry curvature [6–8]; the symmetric partner is the quantum metric [9, 10]. Adiabatic eigenstate defor-

mations are already known to be sensitive probes of integrability breaking [11], and the geometry of parameter space can distinguish ergodic from integrable random-matrix regimes [12]. Most directly, Chen, Colin-Ellerin, Mamroud, and Papadodimas proposed non-Abelian Berry curvature as an intrinsic chaos diagnostic for exactly degenerate BPS microstates [13]. Their contrast is between structured, often vanishing curvature for smooth horizonless sectors and random-matrix-like curvature for supersymmetric black hole microstates.

We test that organizing principle in an exactly degenerate fractional topological manifold with neither supersymmetry nor cohomological protection. The manifold is the kernel of a positive semidefinite bosonic fractional-quantum-Hall parent Hamiltonian. Its degeneracy is exact and algebraic, and its projector is separated from the complement by a gap. This setting isolates the proposed geometric diagnostic from the special BPS mechanism and makes three questions answerable:

1. Does a curvature ramp survive when the energy form factor is exactly silent?
2. Does that ramp distinguish random local tangents from structured tangents at the same active rank?
3. Can energy chaos and projector-geometric chaos be varied independently?

Our answer to all three questions is yes, with important qualifications. The metric-normalized curvature is a compression of a fixed signature matrix by the row space of projector-to-complement response channels. A

Haar row space gives an exact finite-dimensional complex Jacobi point process, allowing us to derive its connected form factor without an asymptotic or empirical fit. The physical random-local-tangent ensemble follows this prediction over the registered two-level window. In contrast, a complete momentum-resolved tangent family has full active rank but only a small number of distinct curvature eigenvalues and fails a three-part falsification test: adjacent-gap repulsion, connected form factor, and number variance.

The central evidence is causal rather than correlational. We vary the Hamiltonian restricted to the degenerate fiber, PHP , from Poisson to GUE statistics while holding P and the projector-to-complement channels fixed; the energy spectrum changes, whereas the Berry curvature does not. Separately, we interpolate the tangent channels from Fourier structured to random local directions at fixed parent Hamiltonian. Local Jacobi repulsion appears at a smaller interpolation strength than the full registered form-factor window, while long-range number variance retains microscopic memory. The resulting order

$$\begin{aligned} \text{local repulsion} &\longrightarrow \text{two-level ramp} \\ &\longrightarrow \text{long-range rigidity} \end{aligned} \quad (2)$$

is a measured hierarchy of geometric correlations. The form-factor variable is a dimensionless Fourier scale of unfolded curvature eigenvalues; it is not physical time and is not assigned a dynamical Thouless interpretation.

Finally, an independent rank sequence through $D = 800$ exposes a finite-dimensional effect invisible in standard full-rank ensembles. When the active rank exceeds the dimension M of either signature polarization, exact atoms appear at $\lambda = \pm 1$. We prove that these deterministic modes reduce the connected plateau from one to k/D , where $k = 2M - D$ is the continuous-sector dimension, while the continuous complement retains the Jacobi ramp. This separates exact geometric structure from nonintegrable correlation structure within one curvature matrix.

The paper is organized as follows. Section II derives the projector response, signature compression, finite-Jacobi kernel, and atom theorem. Section III introduces the fractional topological manifold and the structured falsification control. Section IV performs the independent spectral and geometric interventions. Section V gives the finite-rank atom form factors. Section VI resolves the correlation hierarchy and its statistical meaning. Appendix A records the numerical and inference protocols.

II. PROJECTOR GEOMETRY AND FINITE JACOBI THEORY

A. Gauge-invariant response of a degenerate projector

Let $H(\lambda)$ have an isolated D -dimensional eigenspace at energy E_0 , with spectral projector P , complement

$Q = 1 - P$, and reduced resolvent

$$R = Q[Q(H - E_0)Q]^{-1}Q. \quad (3)$$

For the tangent $V_\mu = \partial_\mu H$, define the projector-to-complement response channel

$$X_\mu = PV_\mu R. \quad (4)$$

In bases of the degenerate fiber and its coupled complement, X_μ is a $D \times M$ matrix. Kato perturbation theory gives, for two real directions v, w ,

$$F_{vw} = i(X_v X_w^\dagger - X_w X_v^\dagger), \quad \Gamma = X_v X_v^\dagger + X_w X_w^\dagger. \quad (5)$$

Here F_{vw} is Hermitian under our convention, and Γ is the positive two-direction quantum metric. Both transform by conjugation under a basis change within P , so the generalized eigenvalues of (F_{vw}, Γ) are gauge invariant.

The raw curvature scale depends on tangent normalization and energy denominators. We remove it by restricting to the $r = \text{rank } \Gamma$ support and defining

$$\Omega = \Gamma^{-1/2} F_{vw} \Gamma^{-1/2}. \quad (6)$$

This normalization does not assume randomness. It converts the diagnostic into the relative orientation of the two response channels. With

$$B = (X_v \ X_w), \quad J = \begin{pmatrix} 0 & iI_M \\ -iI_M & 0 \end{pmatrix}, \quad (7)$$

whitening gives

$$Y = \Gamma^{-1/2} B, \quad YY^\dagger = I_r, \quad \Omega = YJY^\dagger. \quad (8)$$

Thus Ω is a compression of a fixed matrix with M positive and M negative eigenvalues. Its statistics depend only on an r -plane in $\mathbb{C}^{2M}$, i.e., a point on a complex Grassmannian. This observation makes the random-matrix null hypothesis precise: randomness refers to the orientation of the whitened response row space, not to the microscopic Hamiltonian entries.

B. Finite complex-Jacobi point process

If the row space of Y is Haar distributed, the continuously distributed eigenvalues of Eq. (8) form a complex Jacobi ensemble [14, 15]. For $r \leq M$, all $k = r$ levels lie in $(-1, 1)$ and

$$p(\lambda_1, \dots, \lambda_k) \propto \prod_{a < b} |\lambda_a - \lambda_b|^2 \prod_{a=1}^k (1 - \lambda_a^2)^{M-r}. \quad (9)$$

More generally, let

$$k = \min(r, 2M - r), \quad a = |M - r|. \quad (10)$$

After the exact boundary levels derived below are removed, the interior joint density has the same form as Eq. (9) with r replaced by k and exponent a .

Let $p_n^{(a)}(x)$ be orthonormal on $(-1, 1)$ with weight $w_a(x) = (1 - x^2)^a$, and define the weighted projection kernel

$$\mathcal{K}_k(x, y) = \sqrt{w_a(x)w_a(y)} \sum_{n=0}^{k-1} p_n^{(a)}(x)p_n^{(a)}(y). \quad (11)$$

Its diagonal $\rho_k(x) = \mathcal{K}_k(x, x)$ integrates to k . The finite-rank unfolding is therefore

$$u(x) = \int_{-1}^x \rho_k(s) ds. \quad (12)$$

For the unfolded partition sum $Z_F(\tau) = \sum_{j=1}^k e^{-2\pi i \tau u(\lambda_j)}$, we use one normalization throughout:

$$\begin{aligned} K_{F,\text{raw}}(\tau) &= \frac{1}{k} \langle |Z_F(\tau)|^2 \rangle, \\ K_{F,\text{disc}}(\tau) &= \frac{1}{k} |\langle Z_F(\tau) \rangle|^2, \\ K_{F,c}(\tau) &= K_{F,\text{raw}}(\tau) - K_{F,\text{disc}}(\tau). \end{aligned} \quad (13)$$

The determinantal identity $\rho_2(x, y) = \rho_k(x)\rho_k(y) - |\mathcal{K}_k(x, y)|^2$ then yields the exact finite- k prediction

$$K_{J,c}^{(k)}(\tau) = 1 - \frac{1}{k} \int_{-1}^1 dx \int_{-1}^1 dy e^{-2\pi i \tau [u(x) - u(y)]} |\mathcal{K}_k(x, y)|^2. \quad (14)$$

No large- D ramp is inserted by hand. Equation (14) gives $K_{J,c}^{(k)}(0) = 0$ from the projector identity and approaches the unit plateau when the oscillatory cluster term is washed out. We evaluate the double integral by Gauss–Jacobi quadrature; convergence and an independent matrix-beta Monte Carlo comparison are given in Appendix A.

The word “form factor” in Eq. (13) denotes a Fourier transform of the curvature point process. Unlike an energy spectral form factor, τ is not conjugate to physical time. The ramp is evidence for two-level Jacobi correlations and nothing in this definition alone implies operator growth or thermalization dynamics.

C. Boundary-atom theorem and form-factor plateau

When $r > M$, the row space $\mathcal{S} \subset \mathbb{C}^{2M}$ must intersect both M -dimensional eigenspaces $\mathcal{H}_{\pm}$ of J :

$$\dim(\mathcal{S} \cap \mathcal{H}_{\pm}) \geq r + M - 2M = r - M. \quad (15)$$

Every vector in these intersections is an exact eigenvector of the compression with eigenvalue ± 1 . For a generic r -plane the bound is saturated, giving

$$m_+ = m_- = r - M, \quad k = 2M - r. \quad (16)$$

The atoms are deterministic under the Jacobi ensemble, so they contribute no connected fluctuation. If the full

spectrum is normalized by its total rank r , the exact connected form factor is consequently

$$K_{J,c}^{\text{full}}(\tau) = \frac{k}{r} K_{J,c}^{(k)}(\tau), \quad K_{J,c}^{\text{full}}(\infty) = \frac{k}{r}. \quad (17)$$

This is a finite-dimensional theorem rather than a fitted plateau.

For completeness, the raw, non-unfolded form factor retains deterministic interference with the atoms. Writing

$$\begin{aligned} Z_{\text{full}}(s) &= Z_{\text{at}}(s) + Z_{\text{int}}(s), \\ Z_{\text{at}}(s) &= m_- e^{2\pi i s} + m_+ e^{-2\pi i s}, \end{aligned} \quad (18)$$

gives the exact decomposition

$$K_{\text{raw}}^{\text{full}} = \frac{|Z_{\text{at}}|^2}{r} + \frac{2 \text{Re}[Z_{\text{at}}^* \langle Z_{\text{int}} \rangle]}{r} + \frac{\langle |Z_{\text{int}}|^2 \rangle}{r}. \quad (19)$$

Equations (17) and (19) distinguish deterministic geometry from correlated fluctuations and provide two independent numerical closure tests.

III. ENERGY-SILENT FRACTIONAL TOPOLOGICAL MANIFOLD

A. Exact zero modes and tangent families

Our physical anchor is a bosonic Kapit–Mueller lattice parent projected to its exactly flat lowest band [16]. At particle number $N = 3$ and flux number $n = 10$, the projected contact interaction is positive semidefinite and has an isolated zero-mode manifold of dimension $D = 50$. The complement reached by local potentials has dimension $M = 170$ per tangent direction. The zero modes are lattice Laughlin quasihole states obeying the cyclic $(1, 2)$ exclusion rule [17–19]. The numerical kernel width is 5.78×10^{-16} , whereas the external gap is 0.106672.

This is an exact flat band but not a supersymmetric one. There is no supercharge, Witten index, BPS shortening condition, or cohomological protection. The exact kernel follows from the frustration-free parent construction. The open external gap makes P , R , and F_{vw} well defined, while the exact internal degeneracy enforces Eq. (1).

We compare two local-potential tangent families at this same parent:

$$V[v] = \sum_{\mathbf{x}} v_{\mathbf{x}} \hat{n}_{\mathbf{x}}, \quad \sum_{\mathbf{x}} v_{\mathbf{x}} = 0. \quad (20)$$

The physical ensemble contains 20000 independent random tangent pairs, divided before analysis into 12000/4000/4000 training, validation, and test matrices. Every result below uses the untouched 4000-matrix test set.

The negative control is the complete nonzero Fourier family on the 5×5 lattice. For momentum $\mathbf{q} \neq 0$, take

$$v_{\mathbf{x}}^{(\mathbf{q})} \propto \cos(\mathbf{q} \cdot \mathbf{x}), \quad w_{\mathbf{x}}^{(\mathbf{q})} \propto \sin(\mathbf{q} \cdot \mathbf{x}), \quad (21)$$

normalized in the same many-body tangent metric as the random potentials. All 24 nonzero momenta are enumerated, but $\mathbf{q}$ and $-\mathbf{q}$ form only 12 independent orbits and are never counted as independent samples. Each Fourier pair has the same active rank $r = 50$ as a random pair, yet its curvature has at most ten distinct eigenvalues. It therefore rules out explanations based only on rank, gap, parent Hamiltonian, or normalization.

B. Spectral silence and a geometric ramp

Figure 1 states the main contrast. The exact energy form factor follows Eq. (1): the raw curve is $D = 50$, and the connected curve is 0 for every t . The curvature spectrum, however, distinguishes structured Fourier tangents from random local tangents at the same rank.

At $\tau = 0.5$, the connected curvature form factors are

$$\begin{aligned} K_{F,c}^{\text{structured}} &= 4.476, \\ K_{F,c}^{\text{physical}} &= 0.502, \quad K_{J,c}^{(50)} = 0.501. \end{aligned} \quad (22)$$

The physical value is thus close to the parameter-free finite-rank prediction, whereas the structured value is almost an order of magnitude larger. The agreement is assessed on the whole registered interval with simultaneous bands, not from the single value in Eq. (22).

C. Three-scale falsification

A ramp alone is not a sufficient chaos diagnostic: smooth densities, degeneracies, or an inconsistent connected subtraction can produce misleading Fourier features. We therefore require agreement at three resolutions of the same point process:

$$P(r) \longleftrightarrow K_{F,c}(\tau) \longleftrightarrow \Sigma^2(L), \quad (23)$$

testing nearest-neighbor repulsion, two-level correlations across Fourier scales, and long-range counting fluctuations, respectively. Related uses of non-energy spectra emphasize that a ramp probes eigenvalue correlations rather than the physical identity of the operator [20].

Figure 2 rejects two overly strong readings. First, full rank does not imply geometric chaos: the Fourier control has rank 50 and fails all three tests. Second, the physical curvature is not literally a Haar matrix at every scale. Its one-point density has held-out L^1 distance 0.2910 from Haar; a covariance-deformed row-space model reduces this to 0.1237, a 57.5% improvement. At long range,

$$\Sigma_{\text{phys}}^2(8) - \Sigma_{\text{Haar}}^2(8) = 0.12582 \in [0.11428, 0.13736]. \quad (24)$$

The supported statement is therefore correlation-scale dependent: Jacobi local repulsion and a finite- D ramp coexist with non-Haar one-point structure and long-range memory.

IV. INDEPENDENT SPECTRAL AND GEOMETRIC INTERVENTIONS

The preceding comparison could still admit a weak interpretation: perhaps the energy and curvature statistics are merely two correlated consequences of a single hidden randomness. We now intervene on the two blocks of a parameter-dependent Hamiltonian separately. The decomposition

$$H = PHP + PHQ + QHP + QHQ \quad (25)$$

shows why this is possible. Internal energy spacings are controlled by PHP , whereas the projector response in Eq. (4) is controlled by $P(\partial_\mu H)Q$, the complement resolvent, and their adjoints. These are logically distinct data.

A. Changing energy statistics at fixed projector

For the spectral intervention we use an effective $D \times D$ fiber Hamiltonian

$$h_\alpha = \sqrt{1 - \alpha^2} h_P + \alpha h_{\text{GUE}}, \quad 0 \leq \alpha \leq 1, \quad (26)$$

where h_P has independent exponential spacings and h_{GUE} is an independent complex Gaussian Hermitian matrix. Both are centered and matched in Hilbert–Schmidt norm before interpolation. We embed $0.1h_\alpha$ in a fixed target block and place a diagonal complement between energies 2 and 3. The external gap never closes, and diagonalization verifies

$$\max_\alpha \|P_\alpha - P_0\|_F = 1.15 \times 10^{-14}. \quad (27)$$

The tangent channels and a reference curvature spectrum are held fixed. This is an effective causal control, not a claim that Eq. (26) is generated by a local microscopic path.

Across 4000 matrices per value of α , the energy adjacent-gap ratio changes from

$$\langle r_E \rangle_{\alpha=0} = 0.385645, \quad \langle r_E \rangle_{\alpha=1} = 0.598677, \quad (28)$$

while the stored curvature spectrum is unchanged to machine precision. Thus conventional spectral chaos can be created without moving the target projector geometry.

B. Changing geometry at fixed exactly degenerate parent

The complementary intervention leaves the Kapit–Mueller parent, its exact zero energy, its projector, and its external gap fixed. For each structured Fourier direction v_q and an independent random local direction η_v , define

$$\begin{aligned} \tilde{v}_g &= \mathcal{N}_G \left[\sqrt{1 - g} v_q + \sqrt{g} \eta_v \right], \\ \tilde{w}_g &= \mathcal{N}_G \left[\sqrt{1 - g} w_q + \sqrt{g} \eta_w \right], \end{aligned} \quad (29)$$

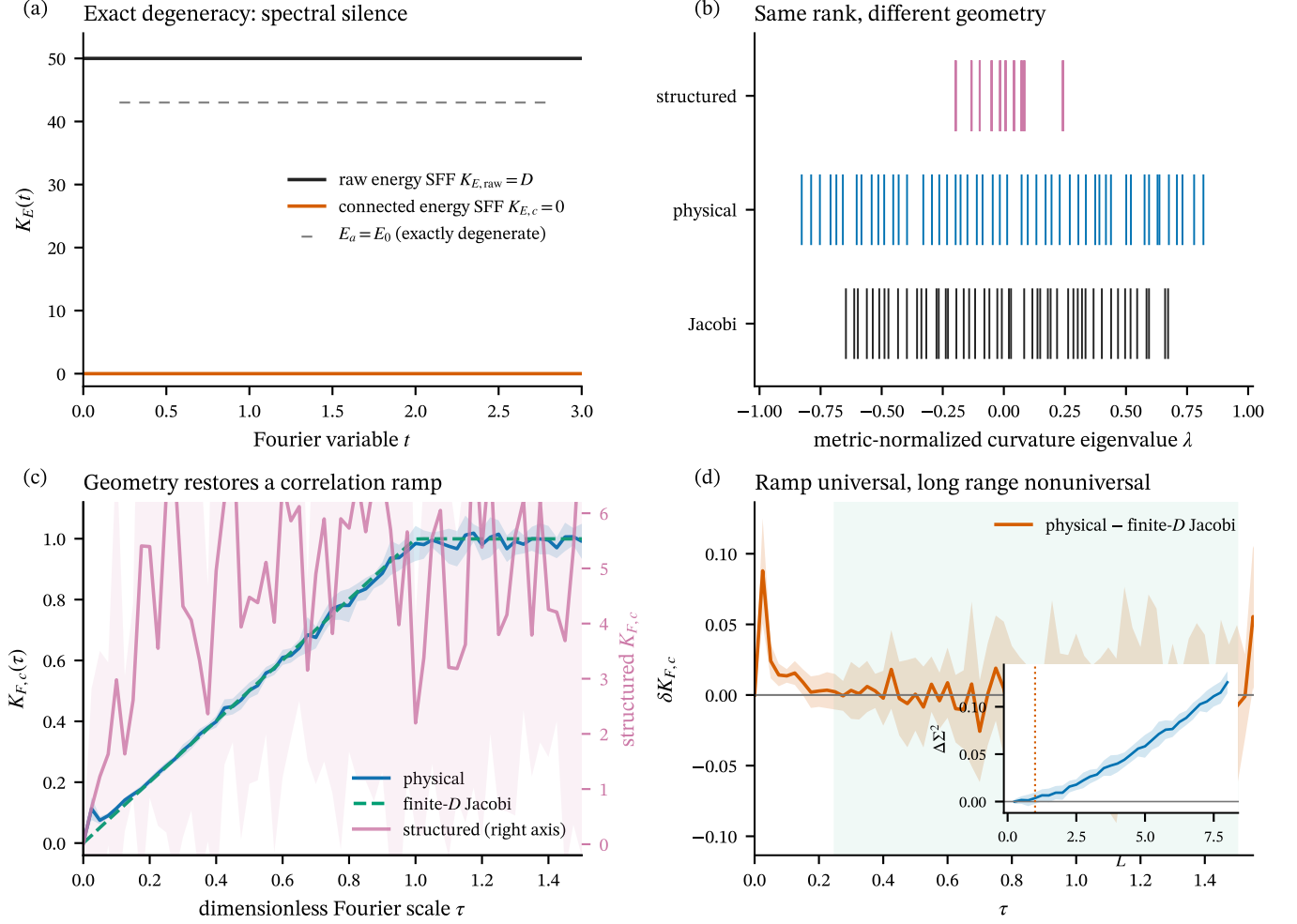


FIG. 1. Exact degeneracy silences the energy spectrum, while projector geometry retains correlated fluctuations. (a) For $D = 50$ exactly degenerate zero modes, the raw energy SFF is the featureless constant D , and the connected SFF vanishes identically. The marks represent coincident energies, not a fluctuating spectrum. (b) Representative spectra of the metric-normalized curvature Ω : structured Fourier tangents retain multiplets, whereas random local tangents produce a resolved spectrum comparable to the Haar–Jacobi reference. (c) Connected curvature SFF of the 4000-matrix physical test ensemble and the exact finite- D Jacobi prediction from Eq. (14). The Fourier control is plotted against a separate right axis because it is parametrically larger. (d) The physical-minus-Jacobi residual has a simultaneous confidence band compatible with zero over the registered interval $0.25 \leq \tau \leq 1.5$, while the inset shows that physical-minus-Haar number variance becomes positive beyond $L \simeq 1$. Here t in (a) is conjugate to energy, whereas τ in (c,d) is only a dimensionless Fourier variable for unfolded curvature eigenvalues.

where $\mathcal{N}_G$ mean-centers and normalizes in the physical tangent Gram matrix. The square-root weights make g a variance fraction before the final normalization. We sample 4000 independent tangent pairs at each positive registered g , maintaining active rank 50 throughout.

Figure 3 gives the operational separation. Changing PHP moves horizontally in panel (d) without changing $P(\partial H)Q$. Changing local tangent structure moves vertically without changing the exactly degenerate parent. Consequently, a Jacobi-like curvature spectrum is not a disguised measurement of an arbitrarily lifted energy spectrum. It diagnoses how the degenerate subspace is embedded in and responds to its surrounding Hilbert

space.

The intervention also supplies an ordering test. The mean curvature gap-ratio interval first overlaps the Haar reference at $g = 0.20$, whereas compatibility with the exact finite- D SFF over a registered nonzero τ interval first occurs at $g = 0.40$. Because these onsets are confidence-set definitions on a discrete grid, they are not critical couplings or universal constants. They demonstrate only that local repulsion is established before the broader two-level ramp under this controlled loss of tangent structure.

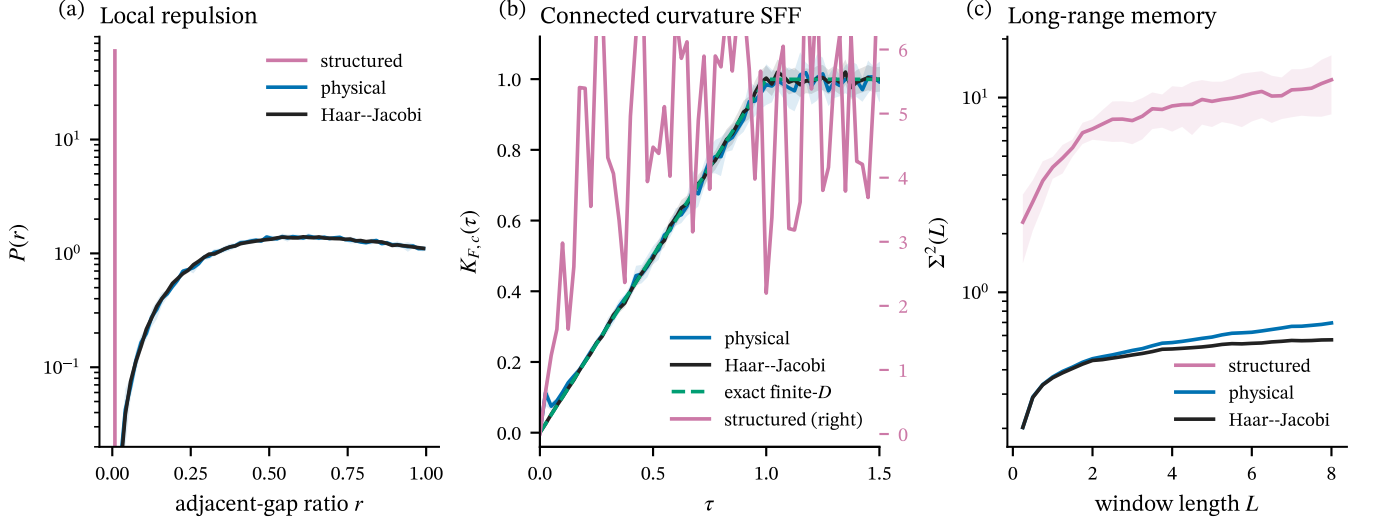


FIG. 2. **Falsification triangle for geometric chaos.** (a) Adjacent-gap-ratio distribution of structured Fourier, physical random-local, and independently sampled Haar-Jacobi curvatures. Exact multiplets make the structured distribution singular near zero; the physical and Jacobi bands overlap. (b) Connected curvature SFF. The empirical Haar curve and the analytic determinantal-kernel result are independent references. (c) Number variance after ensemble-CDF unfolding. Random local tangents have Jacobi rigidity at short windows but a resolved excess at long windows, whereas the structured control is nonuniversal throughout. Bands are simultaneous 95% regions obtained from complete momentum orbits, physical seed blocks, or complete random matrices; eigenvalues within one matrix are never resampled independently.

V. EXACT GEOMETRIC MODES AND THE JACOBI PLATEAU

The $D = 50$ lattice calculation lies in the atom-free regime $D < M$. To test Eq. (17), we use an independent exact root-response construction that reaches both sides of the capacity boundary. For $N = 3$ particles in n orbitals, the cyclic Laughlin zero-mode dimension and one-tangent descendant capacity are

$$D(n) = \frac{n}{n-3} \binom{n-3}{3}, \quad M(n) = 2n(n-3). \quad (30)$$

Two independently sampled one-body response maps are assembled in the monomial root basis and whitened by a thin QR factorization. This directly produces the row isometry in Eq. (8), without using the physical Kapit-Mueller eigenvectors or the $D = 50$ random-tangent data.

The registered sequence contains 8750 matrices at

$$D = 16, 50, 112, 210, 352, 546, 800.$$

It is a finite-rank sequence at fixed particle number $N = 3$, not a thermodynamic limit. At $D = 546$, $M = 540$ and the atom theorem requires six levels at each boundary. At the endpoint $(D, M) = (800, 680)$, it requires 120 levels at each boundary and leaves $k = 560$ continuously distributed levels.

The endpoint plateau is therefore not a numerical shoulder:

$$K_{F,c}^{\text{full}}(\infty) = \frac{2M - D}{D} = 0.7. \quad (31)$$

The reference curve is obtained in two independent ways: evaluation of the determinantal kernel in Eq. (14) and matrix-beta sampling of the Jacobi ensemble. The response calculation uses neither reference to construct its spectra.

This result refines the meaning of random-matrix-like curvature. The full one-point measure at $D > M$ can never converge to a smooth invariant density because a finite fraction of its mass is fixed at ± 1 . Nevertheless, the atom-stripped sector can display Jacobi repulsion and its finite-rank ramp. The exact modes record a dimension-theorem intersection of the accessible row space with the two polarizations of J ; the ramp probes correlations in the remaining orientation. Exact geometry and chaotic geometry are therefore compatible within orthogonal sectors of the same degenerate manifold.

The distinction also protects the analysis from a common error. Removing levels by a numerical distance-to-edge threshold could artificially improve agreement with a smooth reference. Here atom counts are fixed algebraically from (D, M) before any eigenvalues are examined, their numerical positions are audited afterward, and both the full and continuous-sector normalizations are reported.

VI. A HIERARCHY OF GEOMETRIC CORRELATION SCALES

The intervention data allow a scale-resolved statement stronger than a single endpoint comparison. For each g , we subtract the exact finite- D Jacobi curve and test the

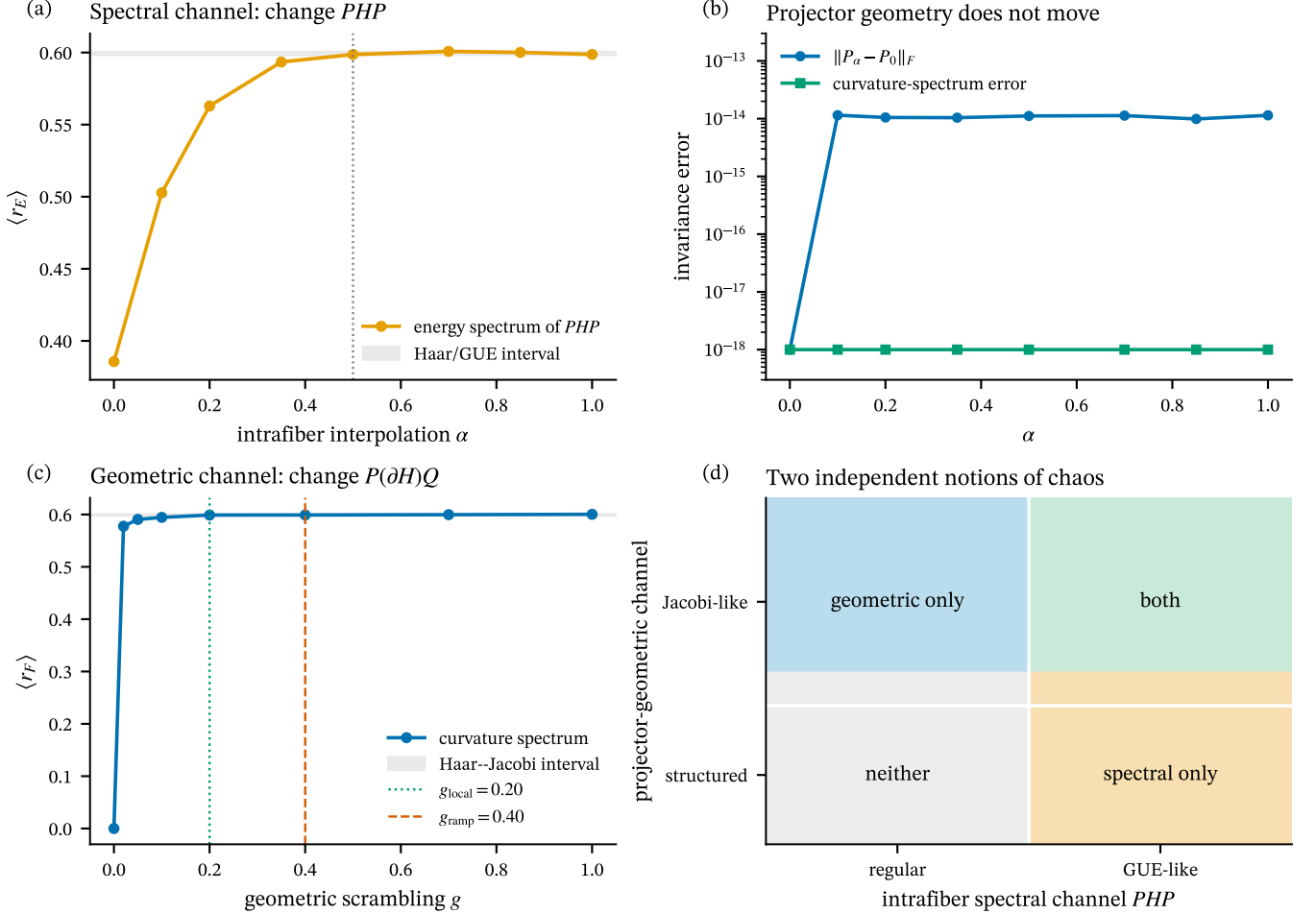


FIG. 3. **Energy chaos and projector-geometric chaos are independent channels.** (a) Mean energy adjacent-gap ratio under the effective fixed-projector interpolation in Eq. (26); it crosses from Poisson to the GUE/Haar interval. (b) The target-projector distance remains at diagonalization roundoff and the supplied curvature spectrum is exactly invariant. (c) At fixed exactly degenerate parent, scrambling only the tangent channels in Eq. (29) changes the curvature gap ratio. Its simultaneous interval first overlaps the Haar–Jacobi interval at $g = 0.20$; the first registered nontrivial SFF window occurs at $g = 0.40$. (d) Logical 2×2 classification. Spectral and geometric chaos can occur separately or together; the original topological flat band occupies the energy-silent/geometrically chaotic quadrant. Bands are 95% simultaneous intervals over the registered interpolation grids.

entire residual process with a simultaneous band. For the physical ensemble we repeat the same procedure using eight independently seeded blocks. Long-range rigidity is tested independently through the number variance

$$\Sigma^2(L) = \left\langle [N_\xi(x, L) - L]^2 \right\rangle, \quad (32)$$

where $N_\xi(x, L)$ counts unfolded curvature levels in a sliding interval of length L .

Three conclusions follow from Fig. 5. First, destroying a small amount of Fourier structure is enough to generate level repulsion, but not enough to reproduce the two-level correlation curve over a nonzero Fourier interval. The registered local and ramp onsets are

$$g_{\text{local}} = 0.20, \quad g_{\text{ramp}} = 0.40. \quad (33)$$

Second, at the physical random-local endpoint the residual band contains zero for every registered $\tau \in [0.25, 1.5]$. This is a curve-level finite- D test; it is more restrictive than matching the value or slope at one point.

Third, two-level agreement does not imply universal long-range rigidity. The physical number variance is compatible with the Haar reference only through $L = 1.0$. At $L = 8$, Eq. (24) excludes zero with a simultaneous interval. The curvature point process therefore loses information in the ordered sequence

$$\begin{aligned} \text{local } P(r) &\prec \text{finite-}D \text{ SFF ramp} \\ &\prec \text{long-range } \Sigma^2(L) \end{aligned} \quad (34)$$

where $A \prec B$ means that A becomes compatible with the invariant reference at a weaker scrambling or shorter correlation scale.

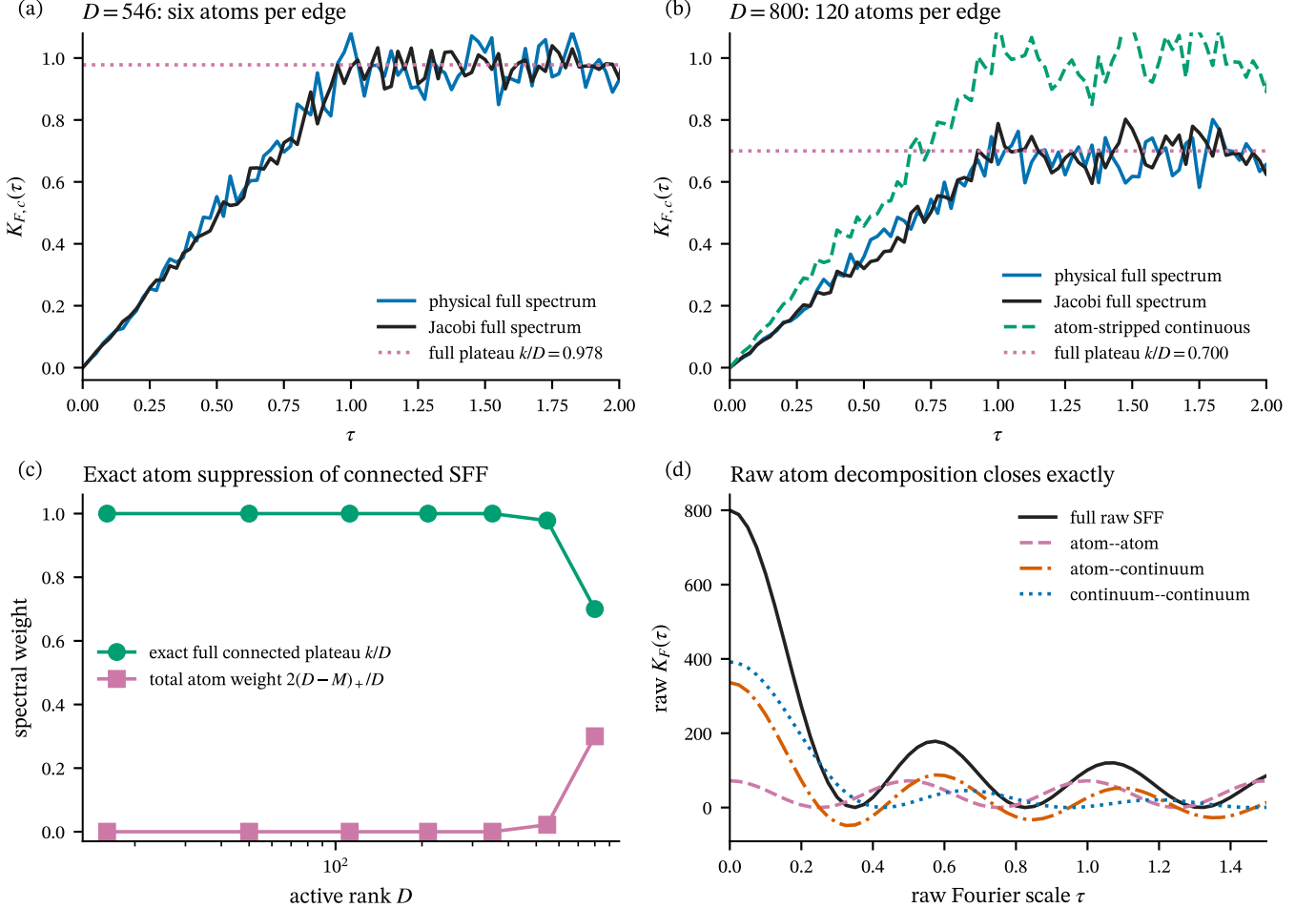


FIG. 4. **Exact boundary atoms suppress the connected form-factor plateau without destroying the continuous Jacobi ramp.** (a) Full-spectrum connected curvature SFF at the first capacity crossing, $D = 546$, $M = 540$, with six exact levels at each of $\lambda = \pm 1$. (b) Endpoint $D = 800$, $M = 680$, with 120 atoms per boundary. The full physical root-response curve follows the full Jacobi prediction, while removing the algebraically identified atoms restores the unit continuous-sector plateau. (c) Across the seven-rank sequence, the exact full connected plateau is k/D ; it falls as the deterministic atom weight $2(D - M)_+/D$ grows. (d) Raw, non-unfolded endpoint SFF separated into atom-atom, atom-continuum, and continuum-continuum terms according to Eq. (19). Their sum closes the full raw form factor to 1.14×10^{-13} .

It is tempting to call Eq. (34) a “geometric Thouless hierarchy.” We use that phrase only as an analogy. There is no real-time diffusion process in Ω , and the present data do not determine a transport time, scrambling time, or conventional Thouless energy. The rigorous content is a hierarchy of confidence-resolved correlation lengths in an unfolded curvature spectrum.

This limitation is also why we do not identify the result with conventional energy-resolved ETH. Our geometric-ETH statement is operational: after removal of a deterministic metric envelope, local and intermediate-range correlations of the projector-response spectrum approach those of a finite invariant point process. The held-out one-point covariance improvement and the surviving number variance show that higher Grassmannian information has not been fully erased. The result is partial geometric ergodicity, not complete Haar randomization.

VII. DISCUSSION

The central result can be stated without analogy:

Exact degeneracy makes the energy spectrum silent, while non-Abelian projector geometry retains a random-matrix correlation ramp whose onset is independent of intramultiplet spectral chaos.

Each clause has a separate test. Equation (1) proves spectral silence. The finite Jacobi kernel gives the geometric null curve. The full-rank Fourier family falsifies the claim that rank or normalization alone produces agreement. The two blockwise interventions show independence. The number-variance residual bounds the range over which universality may be claimed.

This is a sharper statement than “Berry curvature is a

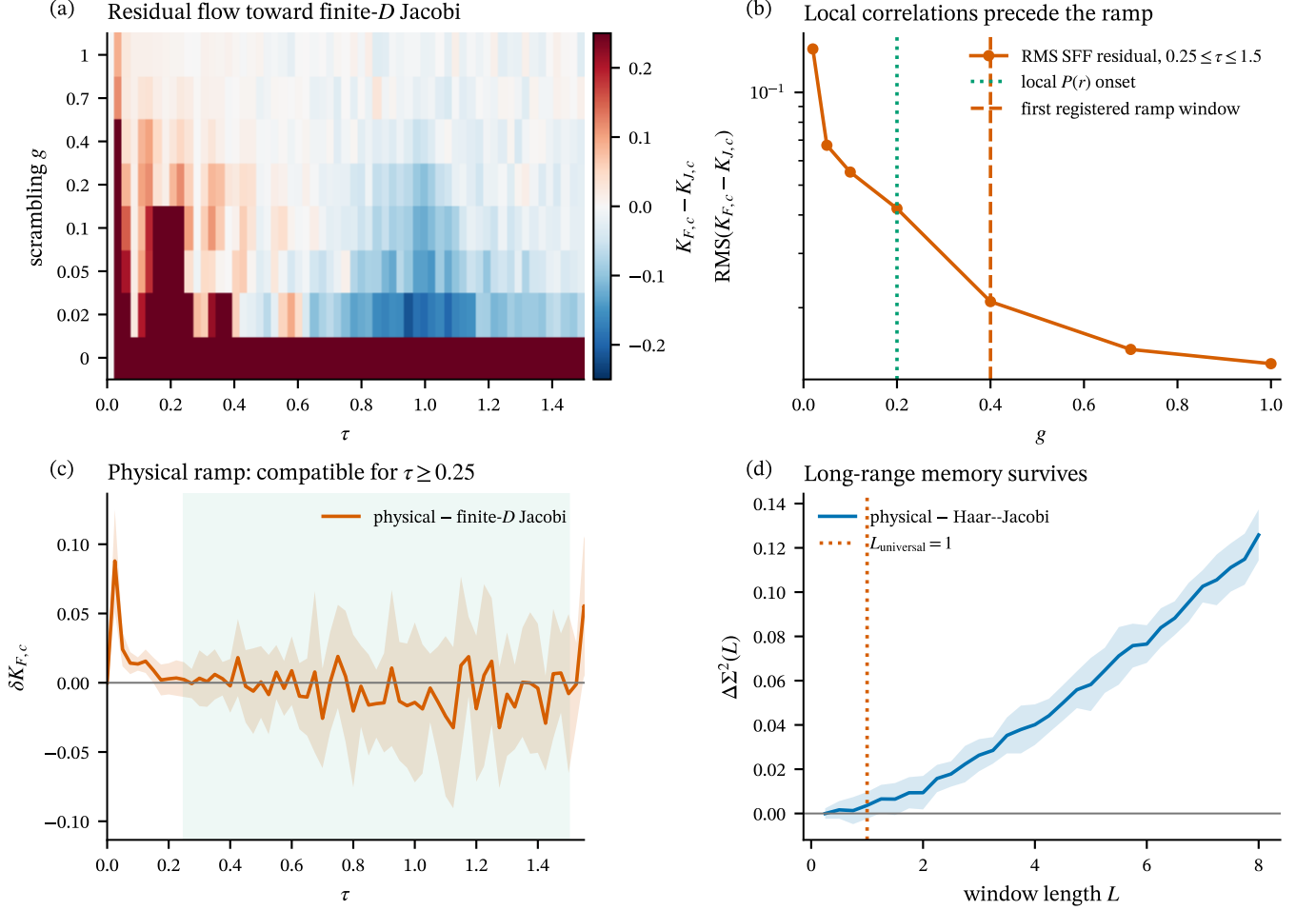


FIG. 5. **Controlled hierarchy from structured to random projector geometry.** (a) Residual connected SFF $K_{F,c}(\tau; g) - K_{J,c}^{(50)}(\tau)$ along the tangent-scrambling interpolation. (b) Root-mean-square residual over the preregistered range $0.25 \leq \tau \leq 1.5$. The local adjacent-gap statistic reaches the Haar interval at $g = 0.20$, while the first nontrivial registered SFF compatibility window occurs at $g = 0.40$. (c) Physical random-local-tangent residual with a simultaneous 95% band; it is compatible with the exact finite- D curve throughout the registered range. (d) Physical-minus-Haar number variance. Short windows are compatible through $L = 1.0$, but the positive long-range residual is resolved thereafter. Onset lines are confidence-set summaries on the sampled grids, not phase transitions.

random matrix.” The physical one-point density remains covariance deformed, long-range counting fluctuations retain microscopic memory, and exact boundary atoms can occupy a macroscopic fraction of the spectrum. What is universal is an ordered subset of correlations. The structured control also shows that a nonzero, full-rank curvature need not be chaotic. Conversely, the atom theorem shows that exact curvature modes need not make the continuous complement integrable.

The relation to Chen *et al.* [13] is direct at the level of the diagnostic and deliberately different at the level of mechanism. Their proposal addresses exactly degenerate BPS sectors and distinguishes supersymmetric black-hole microstates from smooth horizonless geometries; it also exposes nontrivial Chern topology over the $\mathcal{N} = 2$ SYK moduli space. Our system has no SUSY cohomology and no gravitational interpretation. It supplies a condensed-

matter realization in which the following additional pieces can be isolated:

1. a parameter-free finite- D Jacobi prediction for the normalized curvature;
2. a same-parent, same-rank structured negative control;
3. causal separation of PHP from $P(\partial H)Q$; and
4. an exact boundary-atom correction to the connected plateau.

The common principle is that chaos may be encoded in how a degenerate state bundle moves over coupling space even when its energy spectrum carries no internal information.

Several claims remain outside the present calculation. The fixed- $N = 3$ rank sequence is not a thermodynamic scaling law. The fixed-projector Poisson-to-GUE path is an effective block intervention, not a local many-body Hamiltonian interpolation. The curvature SFF is not an out-of-time-order correlator and has no physical time interpretation. We have not computed a Chern number over a closed two-cycle, nor established a smooth-state/black-hole dichotomy. Finally, curve agreement with a Jacobi ensemble is evidence for specified eigenvalue correlations, not a proof that all curvature eigenvectors or all quantum-geometric-tensor components are Haar distributed.

The next analytic step is therefore not to accumulate more unrelated diagnostics. It is to derive the response-channel statistics from locality. Writing a gauge-invariant cumulant expansion for the whitened rows of $PV_\mu Q[Q(H - E_0)Q]^{-1}$ would connect microscopic clustering to the observed order of local, ramp, and long-range scales. A genuine many-body sequence at fixed filling could then test whether the non-Haar covariance and number-variance residual shrink or persist. On the topological side, integrating $\text{Tr } F$ over a controlled coupling two-cycle would determine whether the local correlation hierarchy constrains, or remains independent of, the Chern data emphasized in Ref. [13].

For exactly degenerate quantum matter, the practical prescription is now clear. Do not lift the degeneracy merely to recover level statistics. Compute the isolated projector response, whiten curvature by its quantum metric, compare its continuous spectrum with the exact finite-rank Jacobi process, use structured tangent and fixed-projector controls, and report the correlation scale at which agreement fails. Under that protocol, degeneracy is not an obstruction to diagnosing chaos; it is what makes the geometric channel experimentally and conceptually distinct.

Appendix A: Numerical methods, inference, and reproducibility

1. Physical parent and response cache

The single-particle Kapit–Mueller hopping matrix is constructed with magneto-periodic boundary conditions and projected to its flat lowest band [16]. The many-body contact interaction is represented as a positive constraint factor before diagonalization. Its numerical null space supplies P ; the nonzero eigenvectors and energy denominators supply R in Eq. (3). For a site-potential coefficient vector v , linearity is used to cache $X[v] = PV[v]R$. Every sampled channel is checked against a direct many-body Hilbert–Schmidt norm, and every normalized curvature is audited for $YY^\dagger = I_r$, Hermiticity, rank, and $\text{spec } \Omega \subset [-1, 1]$.

The random-local ensemble has 20000 tangent pairs in eight independently seeded blocks. A fixed permutation determines a 12000/4000/4000 training/validation/test

split before any density, covariance, or form-factor comparison. The 4000-matrix test set is used in all v2 physical figures. The Fourier control enumerates all nonzero momenta, but statistical units are the 12 distinct $\mathbf{q} \leftrightarrow -\mathbf{q}$ orbits.

2. Exact finite-Jacobi quadrature

Equation (14) is evaluated without sampling. At Gauss–Jacobi nodes x_i and weights ω_i of order $q = 512$, we form the weighted orthonormal basis

$$\Phi_{in} = \sqrt{\omega_i} \frac{P_n^{(a,a)}(x_i)}{\sqrt{h_n^{(a)}}}, \quad 0 \leq n < k, \quad (\text{A1})$$

where

$$h_n^{(a)} = \frac{2^{2a+1}}{2n + 2a + 1} \frac{\Gamma(n + a + 1)^2}{\Gamma(n + 1)\Gamma(n + 2a + 1)}. \quad (\text{A2})$$

The discrete projection kernel is $\mathcal{P} = \Phi\Phi^\dagger$, its diagonal is the quadrature mass of ρ_k , and the midpoint cumulative sum of that diagonal implements Eq. (12). The registered $D = 50, M = 170$ calculation has density-mass error 4.19×10^{-13} and basis-orthogonality error 4.18×10^{-13} . Repeating the complete curve at quadrature orders 384 and 512 changes it by less than 5×10^{-5} .

As an independent test, two complex Wishart matrices generate the matrix-beta Jacobi ensemble. For rank 16, 1500 independently sampled spectra agree with the determinantal-kernel connected SFF within 0.075 over the registered test grid. For $r > M$, exact ± 1 atoms are appended algebraically. The identity in Eq. (17) is satisfied to floating-point zero, and the raw decomposition Eq. (19) closes within 1.14×10^{-13} .

3. Interpolation ensembles

The geometric interpolation uses

$$g = 0, 0.02, 0.05, 0.10, 0.20, 0.40, 0.70, 1.00.$$

At every positive g , 4000 independent random local residual pairs are mixed with a registered Fourier pair through Eq. (29). All 28000 positive- g curvature matrices have active rank 50. The complete script-generated scalar adjacent-gap intervals are retained with the numerical artifacts; the Haar–Jacobi simultaneous interval is 0.598440–0.600349.

The spectral interpolation uses

$$\alpha = 0, 0.10, 0.20, 0.35, 0.50, 0.70, 0.85, 1.00$$

and 4000 matrices per value. For every sample, the same Poisson and GUE endpoints are used across α , which reduces interpolation noise without correlating different samples. The target projector is recomputed numerically from the embedded full Hamiltonian rather than assumed fixed. The maximum measured distance in Eq. (27) and the zero curvature-spectrum error are hard delivery gates.

4. Connected estimators and simultaneous inference

Raw, disconnected, and connected form factors are retained separately for every ensemble:

$$\hat{K}_{\text{raw}} = \frac{1}{DN_s} \sum_{s=1}^{N_s} |Z_s|^2, \quad \hat{K}_{\text{disc}} = \frac{1}{D} \left| \frac{1}{N_s} \sum_{s=1}^{N_s} Z_s \right|^2. \quad (\text{A3})$$

Their difference is the displayed connected estimator. This common convention is used for the exact-degenerate energy baseline, physical curvature, Fourier control, interpolation ensembles, and Monte Carlo Jacobi reference. At $\tau = 0$, connected values below 5×10^{-15} are set to zero only after raw-minus-disconnected subtraction.

An eigenvalue is not an independent statistical unit. Physical curves are first averaged within eight independent seed blocks; Fourier curves are averaged within the twelve momentum orbits; the Haar reference uses complete matrices. We construct 10000-replicate Gaussian-multiplier or hierarchical-block processes and take the 95th percentile of the maximum studentized deviation over each registered grid. Thus every shaded region used for an onset or compatibility statement is a simultaneous, not pointwise, confidence band.

For an SFF residual band $[\ell(\tau), u(\tau)]$, the registered compatibility onset is the first grid point $\tau_* \in [0.25, 1.5]$ such that $0 \in [\ell(\tau), u(\tau)]$ for every subsequent registered point. An interpolation value g is said to possess a non-trivial compatibility window only if at least eight grid points remain. The number-variance compatibility extent is the largest L in the contiguous initial interval beginning at $L = 0.25$ whose residual band contains zero. These definitions were fixed before reading the reported onsets.

5. One-point covariance and finite-rank sequence

The v1 analysis retained in the present expanded article tests a distinct observable: the ensemble-mean ambient row-space projector

$$\bar{\Pi} = \mathbb{E}[Y^\dagger Y]. \quad (\text{A4})$$

A covariance-deformed Gaussian row-space ensemble learned from 1024 training matrices and selected on vali-

ation data reduces the held-out density L^1 error from 0.2910 to 0.1237, while preserving the local gap ratio. This result is not used to tune the exact Jacobi SFF in v2; it explains why one-point and long-range observables may retain structure even when local two-level correlations agree.

For the rank sequence, cyclic $(1, 2)$ -admissible root configurations and their one-body descendants are generated independently of the lattice cache. The seven registered sample counts are

2000, 2000, 2000, 1000, 1000, 500, 250.

Atoms are labeled by Eq. (16) before their numerical eigenvalues are inspected. The final $D = 800$ case therefore contains $120 + 120$ exact atoms and 560 interior levels in every matrix. Because $N = 3$ remains fixed, changes along this sequence are described as finite-rank evidence only.

6. Reproducibility gates

The task-local delivery fails if any source artifact hash changes without regeneration; any registered sample count, rank, or interpolation point is missing; training, validation, and test indices overlap; a confidence calculation pools eigenvalues; the finite-Jacobi mass, orthogonality, Monte Carlo, atom, or raw-closure check fails; the fixed projector moves by more than 10^{-12} ; generated figures do not match their input hashes; LaTeX contains undefined references; or any compiled page is omitted from the render audit. The v2 code imports no executable or artifact from another task.

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The numerical artifacts, fixed train-validation-test split, matrix-level uncertainty estimates, and figure-generation scripts are retained with the source of this manuscript.

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